VATSAL KANOJIYA

Python Backend Engineer · Django · PostgreSQL

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SUMMARY

Python backend engineer with nearly 3 years shipping production business systems (billing, ERP, HR, payments) in Python and SQL, and **Django/DRF/PostgreSQL** projects with **concurrency-safe APIs**, Celery jobs and hybrid search. Own features end to end: data model, transactions, **background jobs**, tests and release.

SKILLS

Languages	Python, SQL, JavaScript (Frappe Desk: client scripts, dashboards, custom pages)
Frameworks	Django, Django REST Framework, Frappe / ERPNext v15
Backend	REST APIs, Celery, Redis / RQ, transactions & row locking (select_for_update, advisory locks), idempotency, query optimisation
Data	PostgreSQL (pgvector, full-text search), MySQL / MariaDB, schema design, constraints, indexing
Tooling	Django test framework / unittest, Git, GitHub Actions CI, Docker Compose, Gunicorn, Nginx, Linux, AWS (EC2, S3)
Integrations	Razorpay, WhatsApp Business API, SMS / email gateways, FCM, Zoom, biometric devices

PROJECTS

Super Notes · Django, DRF, PostgreSQL + pgvector, Celery, Redis, Vue 3 github.com/Vatsal-Kanojiya/super_notes

- Notes app that answers questions from your own notes (RAG) with citations; built spec-first with coding agents, every diff reviewed.
- Hybrid retrieval: **pgvector** HNSW similarity and Postgres full-text search merged by **reciprocal rank fusion**, filtered to the owner in SQL, with a recall@5 / MRR evaluation command.
- Multi-device sync with optimistic locking: per-user revisions issued under a **select_for_update** row lock for delta sync; stale edits get a 409 with the server's copy.
- One limits layer for LLM features: per-user quotas under row locks, system-wide caps under Postgres **advisory locks**, and **Idempotency-Key** replay so a retry is never charged twice.

Expense Tracker · Django, DRF, Celery, Redis, PostgreSQL, Docker github.com/Vatsal-Kanojiya/expense_management

- Shared expenses and per-participant splits enforced by database constraints; **N+1** queries removed with select_related / prefetch_related, pinned by **query-count tests**.
- Settle-up safe under concurrent requests (**transaction.atomic** + **select_for_update**); Celery jobs queued via transaction.on_commit; Docker Compose and GitHub Actions CI.

EXPERIENCE

Software Engineer · 360 IT Hub Pvt. Ltd., Bengaluru Feb 2025 – Present

Own delivery across 5–6 concurrent client projects (Python, MariaDB, Redis/RQ on Frappe v15); coordinate a 4-person team spanning backend, Vue web, Flutter mobile and documentation.

- **Sole developer** of a pan-India vending provider's billing and inventory backend, in production: consumption-based billing on multi-site contracts, minimum-order waterfalls, GST invoicing, and **idempotent** approve / cancel / re-approve so a cancel-and-regenerate never double-bills.
- Designed period-level billing for a tax practice (co-built): each bill line tied to the exact service period, server-side guards against **double-billing** and gaps, and a bill-gap report expanding recurring services into billing cycles.
- Re-platformed an undocumented legacy web app after reverse-engineering its schema and behaviour: **~300 API endpoints** behind web and mobile clients, with a client lock plus **SELECT … FOR UPDATE** to stop duplicate record creation.
- Replaced hard-coded vendor calls with a database-configurable **notification router**: two incompatible WhatsApp APIs behind one interface, 12 call sites moved, so switching providers became a settings change.
- **Lead developer** of a reusable HR/attendance app: an absence-scoring engine over shifts, leave and check-ins (bulk-prefetched, 7 queries per range), rule-based shift scheduling with per-row **savepoints**, geofenced and biometric check-in, nightly jobs.
- Build with coding agents (Claude Code) under a **spec-driven process**: pressure-test the design, write the spec with acceptance criteria, slice epics into small tickets, verify against tests and rolled-back transactions, and review every diff, which caught bugs such as quantities above the top slab billing at zero.
- Hardened payment-link flows for expiry, partial settlement and status drift; introduced automated tests to an untested codebase; conducted 50+ technical interviews.

Associate Software Engineer · 360 IT Hub Pvt. Ltd., Bengaluru Dec 2023 – Jan 2025

- Cut report generation from **~5 s to ~1 s** by replacing **N+1** query patterns with batched queries; built payroll reports reconciling attendance, leave and mid-cycle salary revisions.
- Moved bulk result-PDF delivery into **background jobs** uploading to S3, so a mass send of links no longer rendered PDFs on request; OTP-protected public access.

Graduate Apprentice Engineer, B2B Sales · Godrej & Boyce Mfg. Co. Ltd. (Interio), Raipur Jun 2017 – Jun 2018

EDUCATION

M.Tech, Thermal Engineering · RGPV, Bhopal (Trinity Institute of Technology & Research) 2020 – 2023 · CGPA 9.08

B.E., Mechanical Engineering · RGPV, Bhopal (Lakshmi Narain College of Technology) 2013 – 2017 · CGPA 8.27

Certifications Python Full Stack (Django + React), JSpiders, 2023 · Machine Learning, Stanford Online (Coursera), 2022